Learning Representations on Dynamic Graphs

How can models capture evolutional node patterns and predict future links?

2026-01-27
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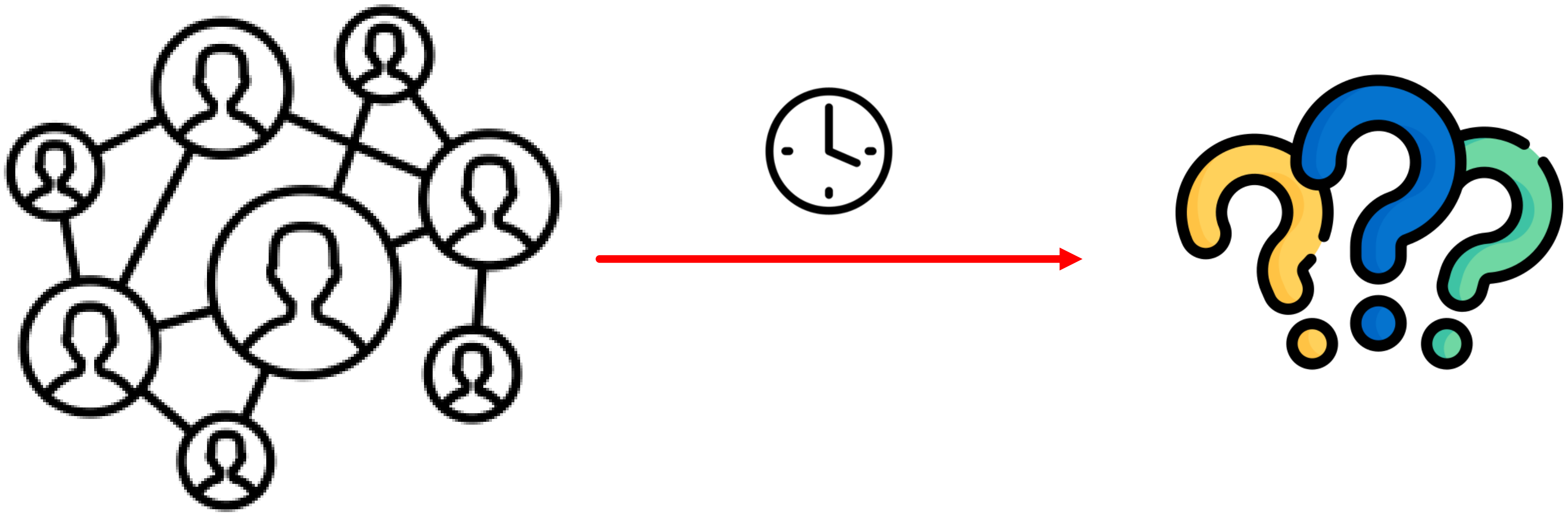
Outline

- ☐ **Introduction**
  - Real-World Networks
  - Limitations of Static Graph Representation
  - Dynamic Graphs
- ☐ **Temporal Random Walks Based - CAWs**
- ☐ **Transformer Based - DyGFormer**

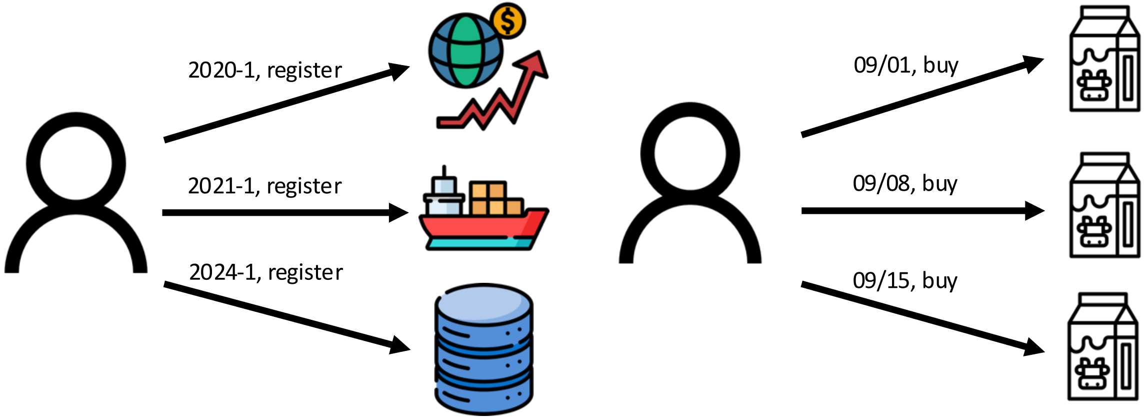
Social Networks

- People as nodes and their relationships as edges
- Will it still be the same over time?



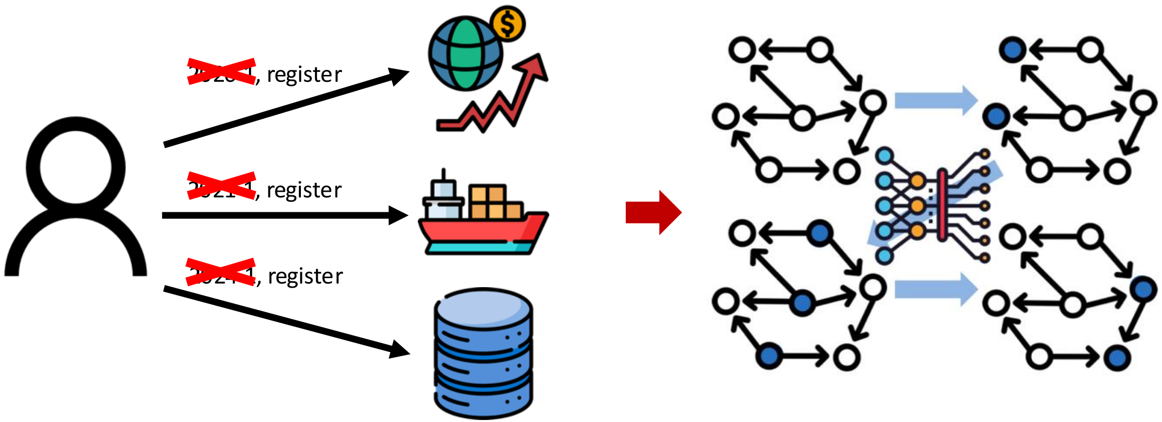
Real-World Interactions

- ☐ Non-static, evolving, and dynamic
- ☐ Crucial insights contained in dynamic structures



Limitations of Static Graph Representation

- ☐ Assumption of a static underlying graph
- ☐ Inability to handle time-evolving graphs
- ☐ Need for alternative modeling approaches

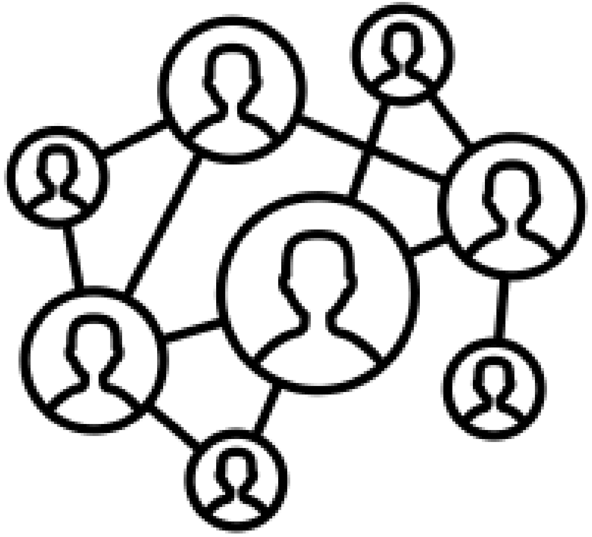


Failure to model temporal dynamics

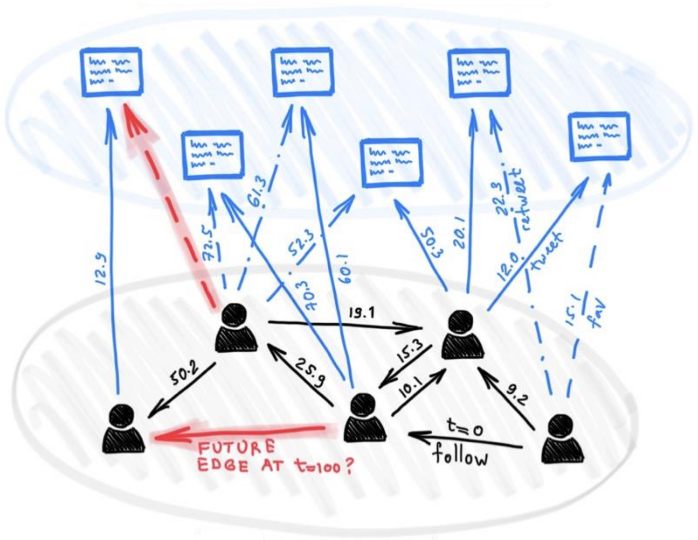
Dynamic Graphs

- Also referred to as temporal graphs, evolving over time

Static Social Networks



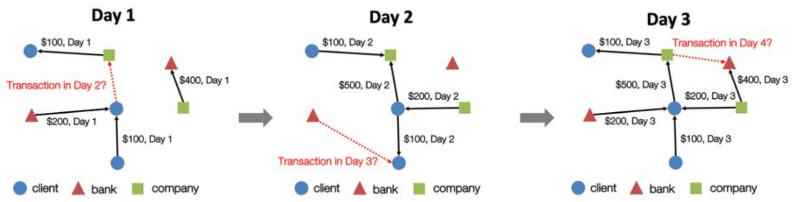
Dynamic Social Networks



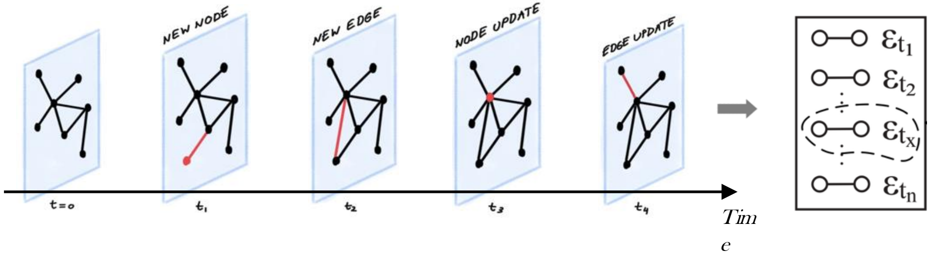
Two Types of Dynamic Graphs

- Discrete-time dynamic graphs (DTDG)
  - A sequence of snapshots aggregated over fixed time gaps
  - Each snapshot as an individual static graph, $G_T = (G_{t_1}, G_{t_2}, ..., G_{t_n})$
- Continuous-time dynamic graphs (CTDG)
  - Event-based graph representation
  - Chronological ordering of graph edges as time-stamped events, $\varepsilon_t = (i, j, t)$

Discrete-Time Dynamic Graph

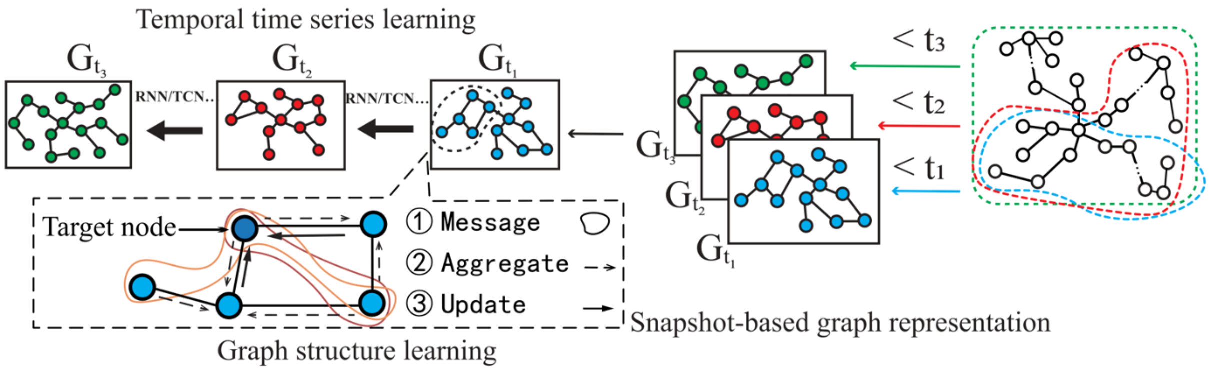


Continuous-Time Dynamic Graph $G_T = (\varepsilon_{t_1}, \varepsilon_{t_2}, ..., \varepsilon_{t_n})$



Processing Discrete-Time Dynamic Graphs

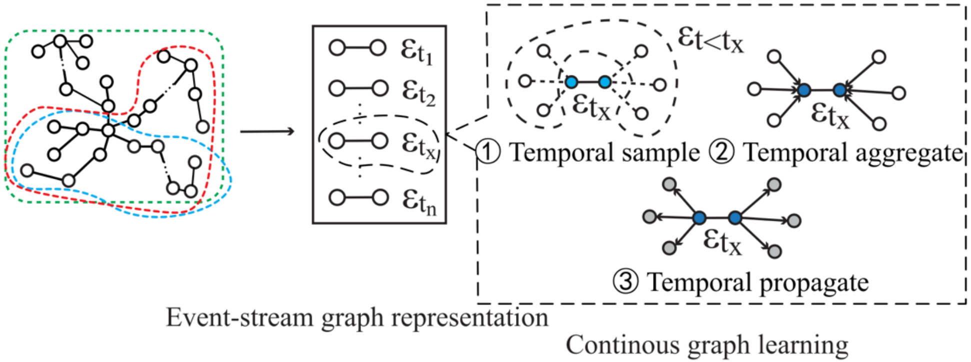
- ☐ Application of static graph modeling methods to each **snapshot**
- ☐ Updating node embeddings or model parameters using each snapshot as input
- ☐ Temporal dependency learning across snapshots using RNNs or similar models
- ☐ Loss of dynamic information in each snap-shot



(a) Discrete-time dynamic graph

Processing Continuous-Time Dynamic Graphs

- Each link as an **event** for the involved nodes
- Learning of the entire graph evolution over continuous time
- Computing node embeddings involved in each interaction using edge-level inputs
- Using RNNs, time-encoding GNNs, and random walks

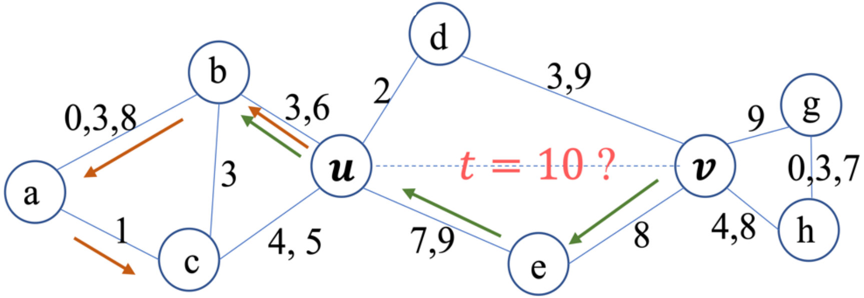


(b) Continous-time dynamic graph

Downstream Tasks

- Link prediction
  - Prediction of whether nodes u and v are connected (interact) at a specific time
- Dynamic node classification
  - Prediction of changes the labels of nodes at given time

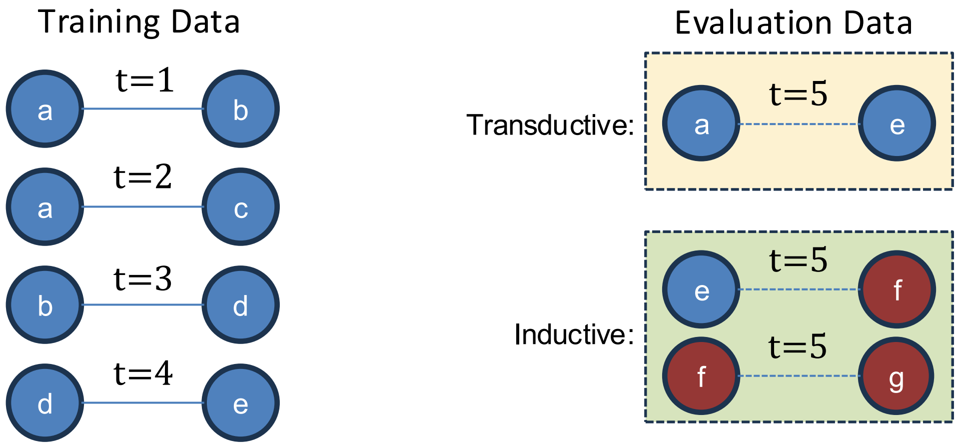
Link Prediction



Dynamic Node Classification

Transductive and Inductive Setting

- Presence of new nodes during evaluation
  - Unseen nodes during training
- Frequent emergence of unseen nodes under continuous event streaming
  - Critical importance of inductive settings in dynamic graph environments
- Learning structure-driven interaction rules beyond node identities



Upcoming Topics

- ☐ CAWs, Causal Anonymous Walks
  - Temporal-random-walk-based modeling of temporal graphs
  - Structural correlation modeling between nodes for link prediction
- ☐ DyGFormer
  - Correlation-aware representation learning for temporal graphs
  - Application of Transformers to dynamic graphs to address limitations of prior methods
- ☐ How can the model capture temporal and structural dynamic information?
- ☐ How can the model guarantee inductiveness?

Inductive Representation Learning in Temporal Networks via Causal Anonymous Walks

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ICLR 2021

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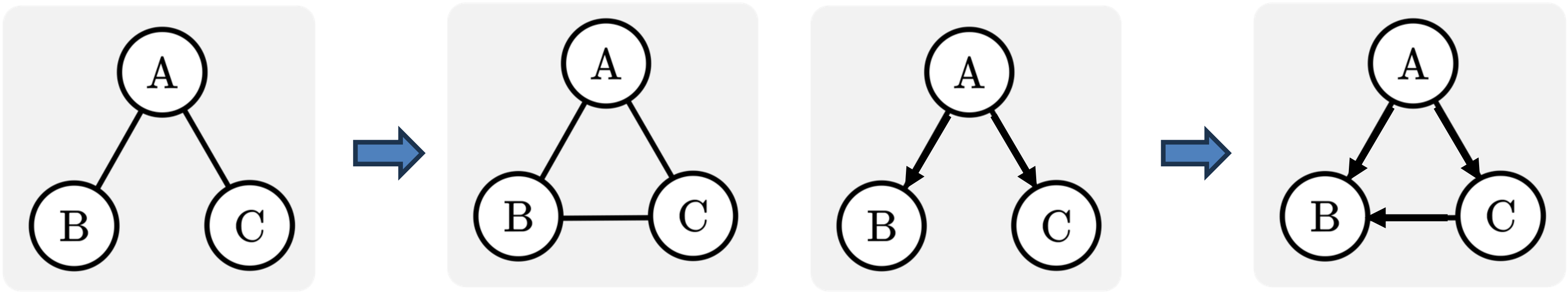
Outline

- ☐ Background
- ☐ Methodology
- ☐ Experiments
- ☐ Conclusion

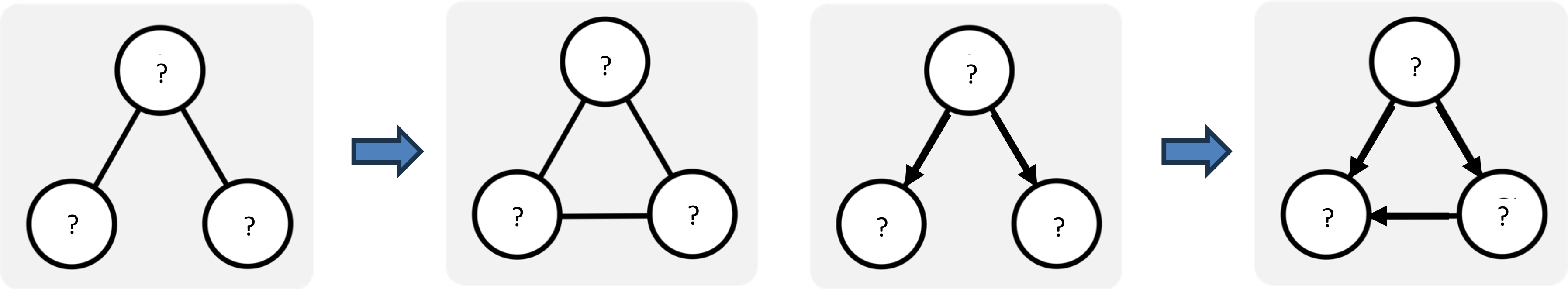
Background

- Insightful Laws on Real-World Systems
  - The triadic closure
    - Two nodes sharing a common node directly connected to them tend to be connected
  - Feed-forward control loops
    - One node activates another node first and then activates a third node
    - Mostly the third node will inhibit the second node later



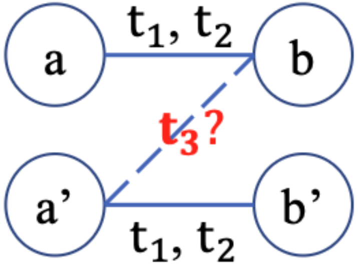
Background

- Inductive Capability
  - Learning graph structures from evolution laws of real-world systems
  - Generalization to systems sharing the same underlying laws
  - No reliance on node identities or an adjacency matrix



Background

- An Ambiguity on Existing Inductive Models
  - Limited to networks with rich link attributes, without capturing structural dynamics
  - Removing node ids and just encoding link timestamps and attributes
    - $\mathbf{Z}(t) = \left[\tilde{\mathbf{h}}_0^{(l-1)}(t) || \Phi_{d_T}(0), \tilde{\mathbf{h}}_1^{(l-1)}(t_1) || \Phi_{d_T}(t - t_1), \ldots, \tilde{\mathbf{h}}_N^{(l-1)}(t_N) || \Phi_{d_T}(t - t_N)\right]^\intercal$
    - Cannot distinguish a vs. a’, b vs. b’
  - TGAT confuses node representations, focusing on structural dynamics only

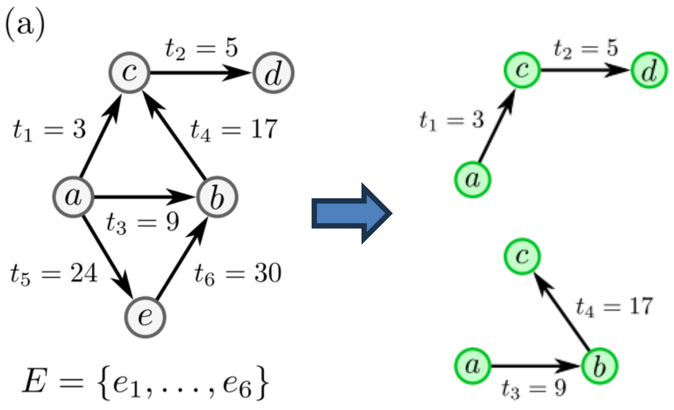


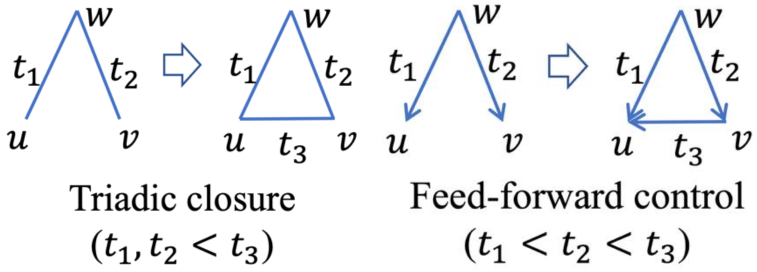
History: $(a, b, \mathrm{t}_1), (a', b', \mathrm{t}_1), (a, b, \mathrm{t}_2), (a', b', \mathrm{t}_2)$;
The order of timestamps: $t_1 < t_2 < t_3$.

Question: Which is more likely to happen,
(a, b, t_3) or (a', b, t_3)?

Background

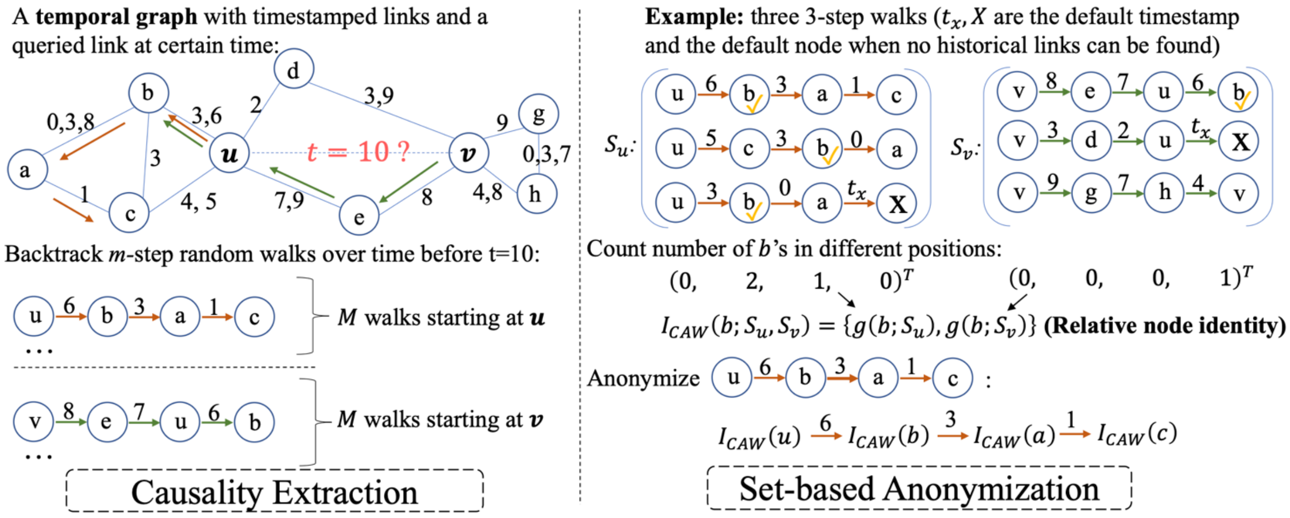
- Temporal Network Motifs
  - Connected subgraphs with links that appear within a restricted time range
  - Essentially reflecting network dynamics
  - Laws of real-world can be viewed as temporal network motifs evolving





Methodology

- Causal Anonymous Walk, CAW
  - A random-walk-based model, extracting temporal motifs
  - Capturing the correlation of temporal motifs, while agnostic to the node id
  - Using set-based anonymization to guarantee inductive learning



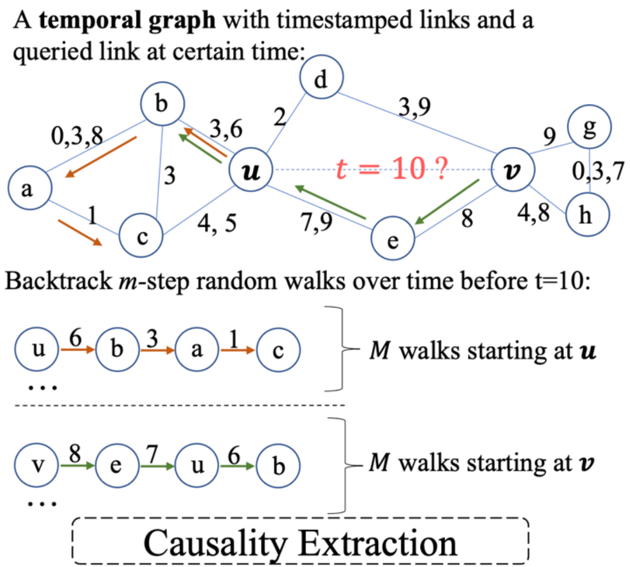
Methodology

Causality Extraction

- Sampling by backtracking the link over time using random walks
- The more recent link, the more informative link
- Hyper-parameter α to sample a link with a probability proportional
  - Large α → emphasizing more on recent links
  - Zero α → uniform sampling

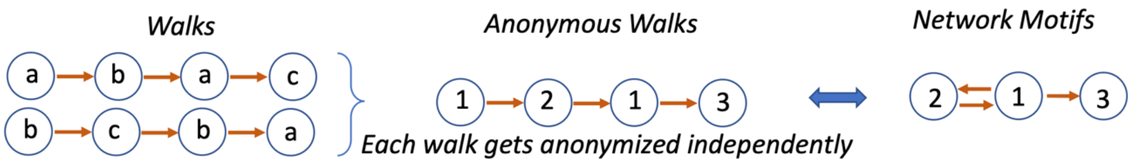
Algorithm 1: Temporal Walk Extraction ($\mathcal{E}$, α, M, m, w_0, t_0)

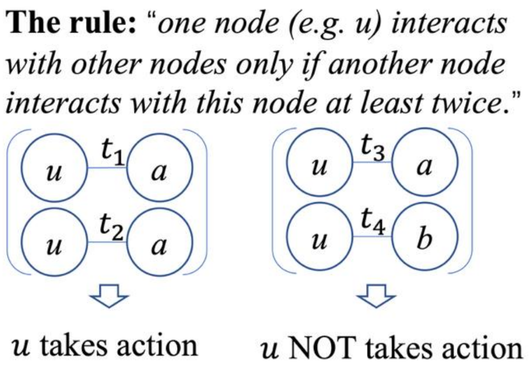
Initialize M walks: $W_i \leftarrow ((w_0, t_0)), 1 \le i \le M$;
for j *from* 1 *to* m **do**
    for i *from* 1 *to* M **do**
        $(w_p, t_p) \leftarrow$ the last (node, time) pair in W_i;
        Sample one $(e, t) \in \mathcal{E}_{w_p, t_p}$ with prob. $\propto \exp\boxed{(\alpha(t - t_p))}$
        Denote $e = \{w', w\}$ and then $W_i \leftarrow W_i \oplus (w', t)$;
Return $\{W_i | 1 \le i \le M\}$;



Methodology

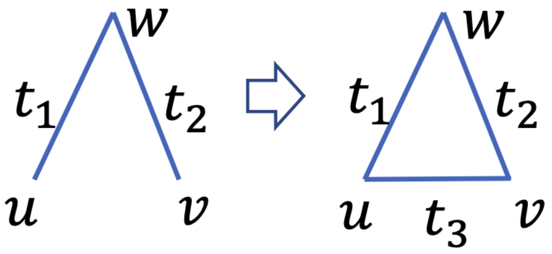
- Anonymous Walks
  - Anonymization step to replace the node ids by the orders of their appearance in each walk
    - $I_{AW}(w;W) \triangleq |\{v_0, v_1, ..., v_{k^*}\}|,$ where k^* is the smallest index s.t. $v_{k^*} = w$.
  - Anonymizing independently within each single walk
    - Node identities are not shared across different walks
    - Losing the correlation network motifs
    - How can the model using AW learn the rule?



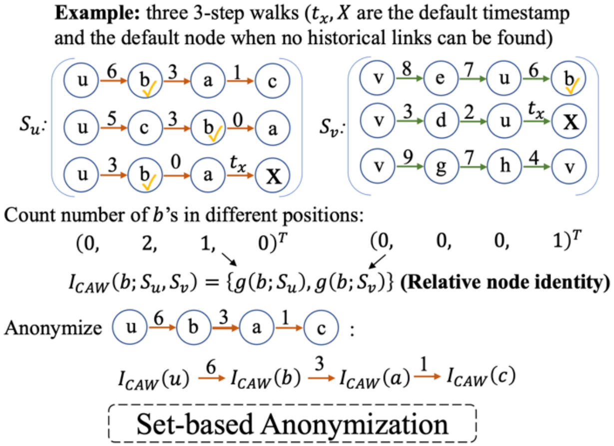


Methodology

- Set-based Anonymization
  - Correlation across different walks could be a key of laws on network dynamics
  - Counting each node's occurrences in different positions within each set
    - Replacing node identities with hit counts to maintain inductiveness
    - $I_{CAW}(w; \{S_u, S_v\}) \triangleq \{g(w, S_u),\ g(w, S_v)\}.$
  - Extracting the correlation between network motifs of each node



Triadic closure
$(t_1, t_2 < t_3)$



Methodology

- Neural Encoding for CAWs
  - $\hat{W} = ((I_{CAW}(w_0), t_0), (I_{CAW}(w_1), t_1), \ldots, (I_{CAW}(w_m), t_m))$.
  - $\text{enc}(\hat{W}) = \text{RNN}(\{f_1(I_{CAW}(w_i)) \oplus f_2(t_{i-1} - t_i)\}_{i=0,1,\ldots,m})$, where $t_{-1} = t_0$,

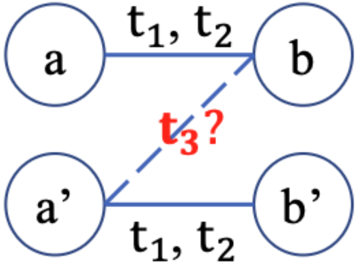
$$f_1(I_{CAW}(w_i)) = \text{MLP}(g(w, S_u)) + \text{MLP}(g(w, S_v)) \oplus f_2(t) = [\cos(\omega_1 t), \sin(\omega_1 t), \ldots, \cos(\omega_d t), \sin(\omega_d t)]$$

$$\Downarrow$$

$$\text{Mean-AGG}(S_u \cup S_v):\ \frac{1}{2M} \sum_{i=1}^{2M} \text{enc}(\hat{W}_i).$$

Methodology

- Relief of Ambiguity on TGAT
  - $ICAW(w; \{S_u, S_v\}) = \{g(w, S_u),\ g(w, S_v)\}$



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The order of timestamps: $t_1 < t_2 < t_3$.

Question: Which is more likely to happen,
(a, b, t_3) or (a', b, t_3)?

$\Downarrow$

$$
\begin{aligned}
ICAW(a; \{S_a, S_b\}) &= \{[2,0],[0,2]\} \\
ICAW(b; \{S_a, S_b\}) &= \{[0,2],[2,0]\}
\end{aligned}
\quad \neq \quad
\begin{aligned}
ICAW(a'; \{S_{a'}, S_b\}) &= \{[2,0],[0,0]\} \\
ICAW(b; \{S_{a'}, S_b\}) &= \{[0,0],[2,0]\}
\end{aligned}
$$

Experiments

Experimental Setup

- Six previous SOTA baselines
  - Snapshot-based methods: DynAERNN, VGRNN, EvolveGNN
  - Event-based methods: JODIE, TGAT, DyRep
- Six real-world public dataset
  - Wikipedia, Reddit, MOOC
  - Social Evolution, Enron, UCI
- Link prediction
  - Transductive
  - Inductive – old vs. new / new vs. new

Experiments

- Performance in AP

Task		Methods	Reddit	Wikipedia	MOOC	Social Evo.	Enron	UCI
Inductive	new v.s. new	DynAERNN	58.63 ± 5.42	54.94 ± 2.29	59.84 ± 1.26	54.76 ± 1.33	54.89 ± 3.79	51.59 ± 3.92
		JODIE	80.03 ± 0.13	76.90 ± 0.49	82.27 ± 0.46†	87.96 ± 0.12†	79.80 ± 1.48†	71.64 ± 0.62
		DyRep	61.28 ± 1.89	57.57 ± 2.56	62.29 ± 2.09	75.42 ± 0.32	69.97 ± 0.92	63.08 ± 7.40
		VGRNN	60.64 ± 0.68	52.55 ± 0.82	65.44 ± 0.82	67.83 ± 0.53	67.93 ± 0.88	67.50 ± 0.92
		EvolveGCN	62.99 ± 0.17	55.64 ± 1.03	52.28 ± 1.80	52.26 ± 1.16	47.36 ± 1.24	80.98 ± 1.09†
		TGAT	95.17 ± 0.91†	93.18 ± 0.73†	72.91 ± 0.92	52.17± 1.94	63.83 ± 3.70	75.27 ± 2.34
		CAW-N-mean	**98.05 ± 0.87***	**96.01 ± 0.25***	**90.36 ± 0.80***	**92.16 ± 1.03***	**93.93 ± 0.66***	**99.63 ± 0.34***
		CAW-N-attn	**98.08 ± 0.66***	**97.86 ± 0.63***	**90.35 ± 0.81***	**93.29 ± 1.90***	**92.73 ± 0.76***	**100.00 ± 0.00***
	new v.s. old	DynAERNN	66.59 ± 2.90	63.76 ± 2.82	82.02 ± 1.59	52.54 ± 0.22	55.50 ± 2.07	57.29 ± 2.52
		JODIE	83.15 ± 0.03	80.54 ± 0.06	87.95 ± 0.08	91.40 ± 0.04†	89.57 ± 0.30	76.34 ± 0.17
		DyRep	66.73 ± 1.99	76.89 ± 0.31	88.25 ± 1.20†	89.41 ± 0.29	95.97 ± 0.28†	93.60 ± 1.47†
		VGRNN	52.84 ± 0.66	60.99 ± 0.55	62.95 ± 0.58	69.20 ± 0.52	67.93 ± 0.88	67.50 ± 0.92
		EvolveGCN	66.29 ± 0.52	53.82 ± 1.64	51.53 ± 0.92	52.01 ± 0.67	46.56 ± 1.89	76.30 ± 0.33
		TGAT	97.09 ± 0.18†	95.17 ± 0.15†	71.77 ± 0.23	52.48 ± 0.52	59.70 ± 1.49	75.01 ± 0.72
		CAW-N-mean	**98.89 ± 0.04***	**99.04 ± 0.04***	**90.99 ± 0.96***	**93.71 ± 0.75***	92.93 ± 0.94	**98.86 ± 0.95***
		CAW-N-attn	**99.90 ± 0.05***	**99.55 ± 0.30***	**91.24 ± 0.93***	**94.17 ± 0.86***	92.38 ± 1.36	**98.53 ± 0.64***
Transductive		DynAERNN	85.58 ± 2.12	76.58 ± 1.41	89.29 ± 0.49	66.58 ± 1.84	60.90 ± 2.70	84.95 ± 2.13
		JODIE	91.14 ± 0.01	91.39 ± 0.04	91.19 ± 0.03†	89.22 ± 0.01	91.94 ± 0.01	80.27 ± 0.08
		DyRep	67.54 ± 2.02	77.36 ± 0.25	90.49 ± 0.03	94.48 ± 0.01†	97.14 ± 0.07†	95.29 ± 0.13†
		VGRNN	50.87 ± 0.81	67.66 ± 0.89	83.70 ± 0.56	78.66 ± 0.67	94.02 ± 0.52	82.23 ± 0.56
		EvolveGCN	54.49 ± 0.73	55.84 ± 0.37	51.80 ± 0.46	56.90 ± 0.54	69.72 ± 0.49	81.63 ± 0.23
		TGAT	98.38 ± 0.01†	96.65 ± 0.06†	69.75 ± 0.23	57.37 ± 1.18	57.37 ± 0.18	60.25 ± 0.31
		CAW-N-mean	**99.96 ± 0.01***	**99.91 ± 0.03***	**92.05 ± 0.88**	**95.90 ± 0.09***	94.93 ± 0.39	**95.89 ± 0.87**
		CAW-N-attn	**99.99 ± 0.01***	**99.89 ± 0.03***	**92.23 ± 0.76***	**96.37 ± 0.09***	96.13 ± 0.37	**98.86 ± 0.38***

Experiments

- Ablation Study with CAW-N-mean

No.	Ablation	Wikipedia	UCI	Social Evo.
1.	original method (CAW-N-mean)	98.49 ± 0.38	99.12 ± 0.33	94.54 ± 0.69
2.	remove $f_1(I_{CAW})$	96.28 ± 0.66	79.45 ± 0.71	53.69 ± 0.21
3.	remove $f_2(t)$	97.99 ± 0.13	95.00 ± 0.42	71.93 ± 2.32
4.	remove $f_1(I_{CAW}), f_2(t)$	88.94 ± 0.85	50.01 ± 0.02	50.00 ± 0.00
5.	replace $f_1(I_{CAW})$ by one-hot(I_{AW})	96.55 ± 0.21	85.59 ± 0.34	68.47 ± 0.86
6.	fix $\alpha = 0$	75.10 ± 3.12	87.13 ± 0.49	78.01 ± 0.69

Experiments

- A Drawback of CAWs

	Reddit	Wiki	MOOC	LastFM	GDELT
JODIE	5 sec	2 sec	4 sec	11 sec	16 sec
DySAT	33 sec	6 sec	16 sec	41 sec	83 sec
TGAT	15 sec	4 sec	8 sec	28 sec	41 sec
TGN	8 sec	2 sec	5 sec	15 sec	32 sec
CAWs-mean	1, 893 sec	277 sec	641 sec	1, 797 sec	4, 544 sec
CAWs-attn	1, 930 sec	282 sec	653 sec	1, 832 sec	4, 634 sec
TGSRec	538 sec	157 sec	656 sec	1, 810 sec	3, 707 sec
APAN	13 sec	4 sec	9 sec	28 sec	25 sec
GraphMixer	12 sec	3 sec	7 sec	21 sec	32 sec

Conclusion

- Existing Models
  - Non-inductive designs via node identities
  - Inductive ambiguity under structural dynamics

- CAWs for Inductive Temporal Link Prediction
  - Set-based anonymization for identity-agnostic yet structure-aware representations
  - Modeling correlations across walks/motifs through shared-node hit-count patterns

- Robustness on Various Dataset
  - Robust Performance without node and link Features

Towards Better Dynamic Graph Learning: New Architecture and Unified Library

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NeurIPS 2023

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Outline

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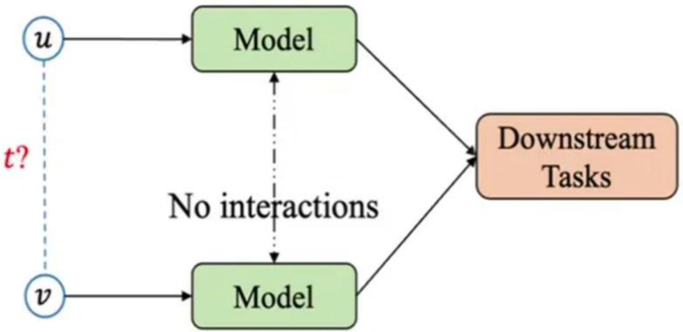
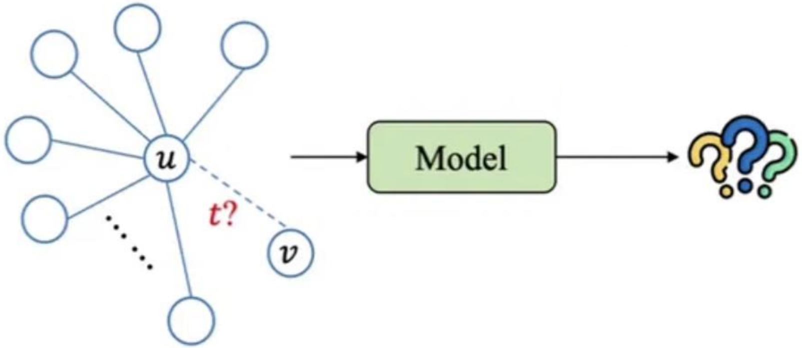
- ☐ Methodology
- ☐ Experiments
- ☐ Conclusion

Background

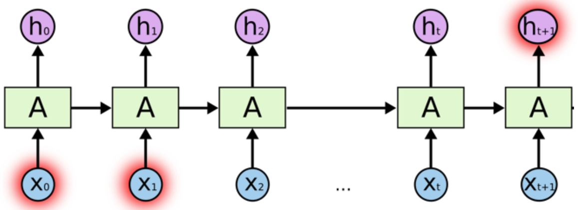
- Two Major Issues
  - Separate learning process
    - Computing independently temporal representations of nodes
    - Ignoring nodes' correlations
  - Only working for nodes with fewer interactions
    - Sampling strategy to truncate the interactions for feasible calculations
  - Lack the ability to capture either nodes' correlations or long-term temporal dependencies

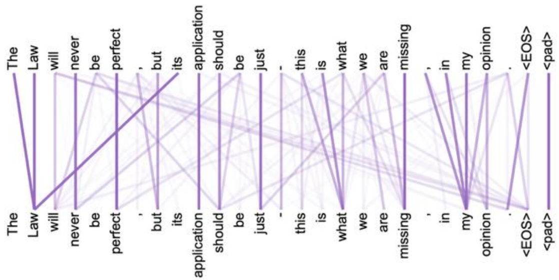


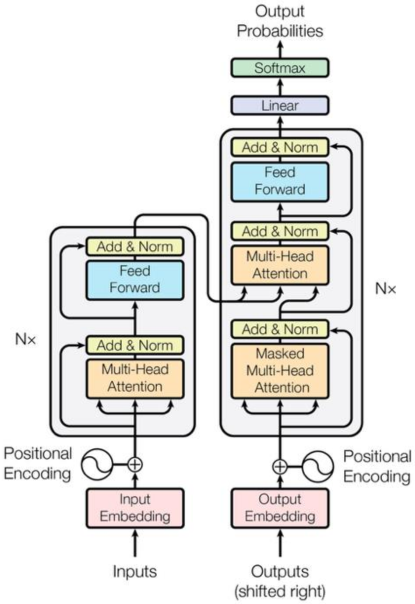


Background

- ☐ Recall the Transformer
  - ■ Entirely relied on self-attention mechanism
  - ■ Not using recurrence or hidden states
    - ☐ Naturally avoid vanishing or exploding gradients
  - ■ Ability to capture long-term dependencies



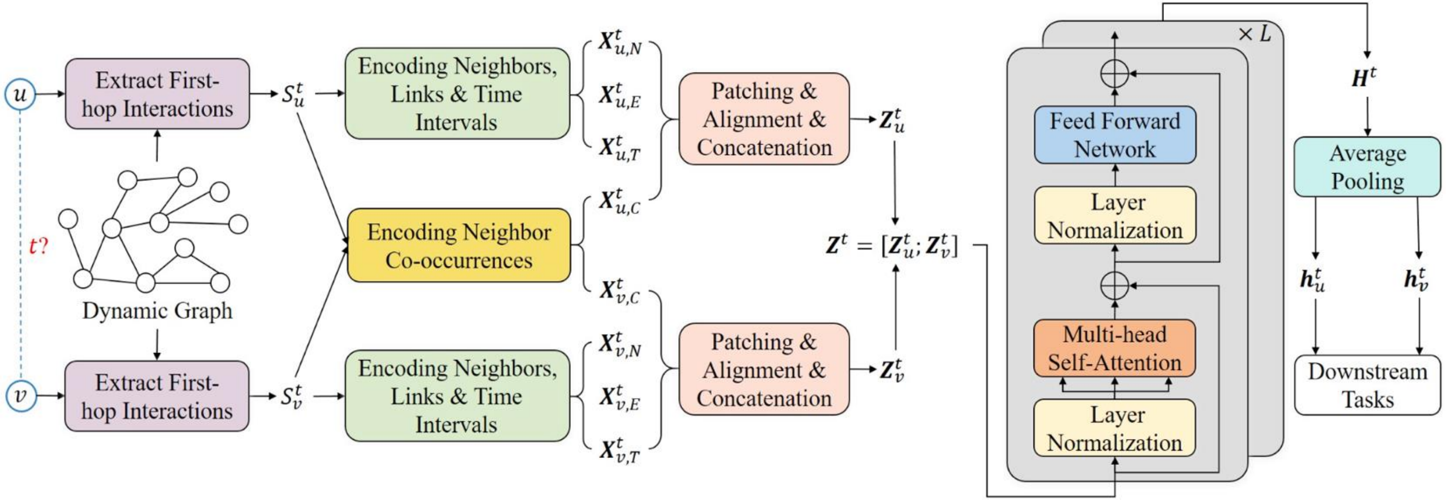




Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In Advances in Neural Information Processing Systems, 2017

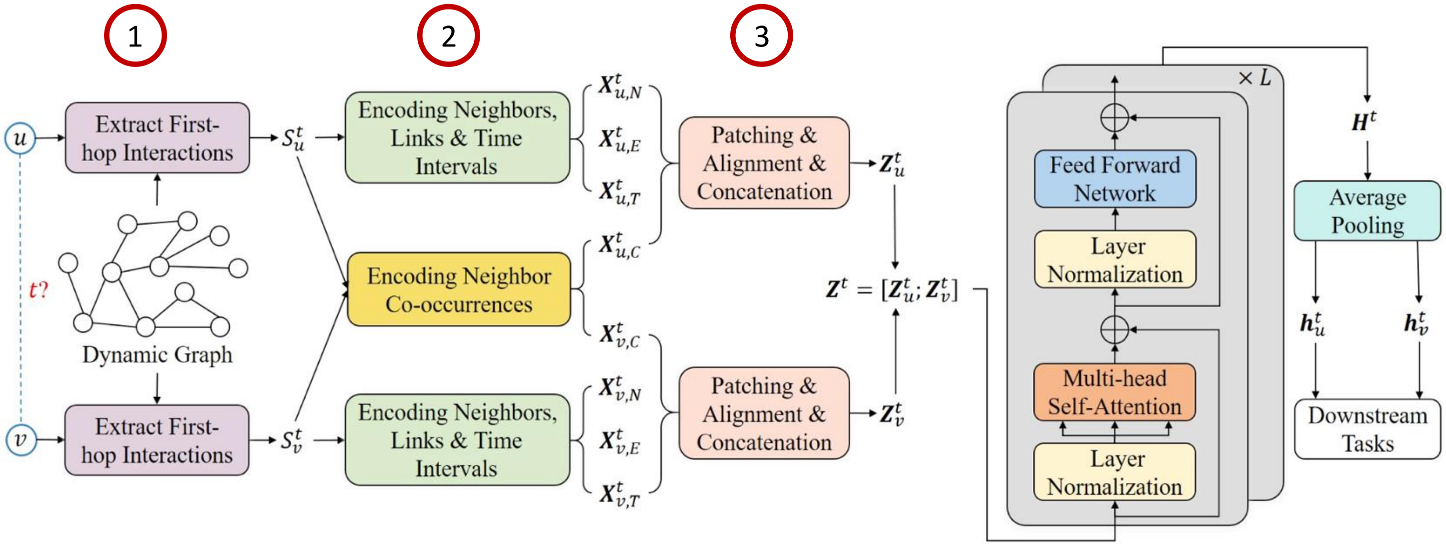
Methodology

- DyGFormer
  - A new Transformer-based dynamic graph learning architecture
  - Learning from the sequences of nodes' historical first-hop interactions
    - Exploiting the correlations of nodes in each interaction
  - Benefit from longer histories via preserving local temporal proximities
    - Efficiently capturing long-term temporal dependencies



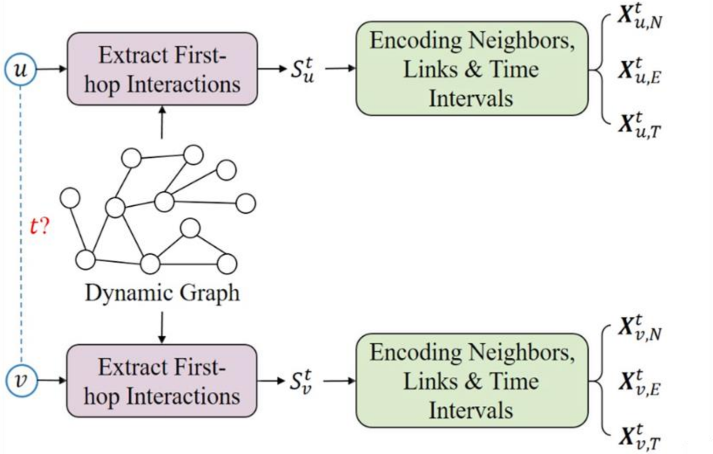
Methodology

- Overall Process of DyGFormer
  - Extracting historical first-hop interactions of each node before time t
  - Encodings of neighbors, links, time intervals, and neighbor co-occurrences for each sequence
  - Dividing each encoding into multiple patches and feeding patches into Transformer



Methodology

- Encoding Process
  - Only learning from the sequences of nodes' historical first-hop interactions
    - $\mathcal{S}_u^t = \{(u, u', t')\,|t' < t\} \cup \{(u', u, t')\,|t' < t\}$
  - Retrieving the features of involved neighbors and links in $\mathcal{S}_u^t$
    - $\boldsymbol{X}_{u,N}^t \in \mathbb{R}^{|\mathcal{S}_u^t| \times d_N}$ and $\boldsymbol{X}_{u,E}^t \in \mathbb{R}^{|\mathcal{S}_u^t| \times d_E}$
  - Learning the periodic temporal patterns by encoding the time interval
    - $\Delta t' = t - t'$ via $\sqrt{\frac{1}{d_T}}\left[\cos\left(w_1 \Delta t'\right), \sin\left(w_1 \Delta t'\right), \cdots, \cos\left(w_{d_T} \Delta t'\right), \sin\left(w_{d_T} \Delta t'\right)\right]$
    - $\boldsymbol{X}_{u,T}^t \in \mathbb{R}^{|\mathcal{S}_u^t| \times d_T}$



Methodology

☐ Neighbor Co-occurrence Encoding

- The appearing frequency of a neighbor in a sequence indicates its importance
- The co-occurrence of a neighbor in each sequence reflecting the correlations between nodes
- More common historical neighbors between u and v → higher likelihood of future interaction

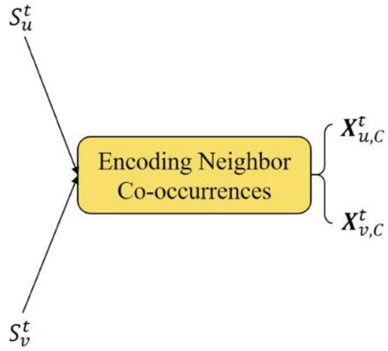
Historical neighbors of u → $\{a, b, a\}$

Neighbor co-occurrence features of u → $\boldsymbol{C}_u^t = [[2,1],[1,2],[2,1]]^\top$

$$\boldsymbol{X}_{*,C}^t = f\left(\boldsymbol{C}_*^t[:,0]\right) + f\left(\boldsymbol{C}_*^t[:,1]\right) \in \mathbb{R}^{|\mathcal{S}_*^t| \times d_C}$$

Historical neighbors of v → $\{b, b, a, c\}$

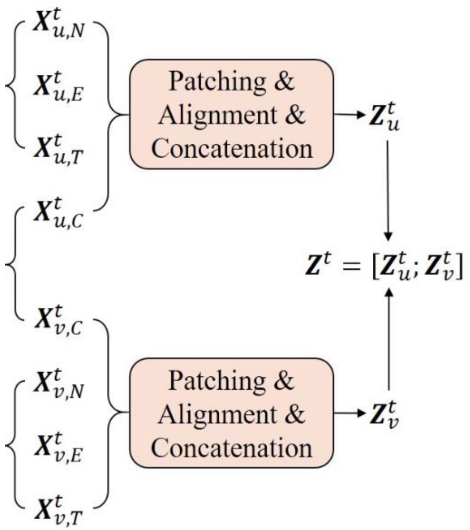
Neighbor co-occurrence features of v → $\boldsymbol{C}_v^t = [[1,2],[1,2],[2,1],[0,1]]^\top$



Methodology

Patching and Employing a Transformer

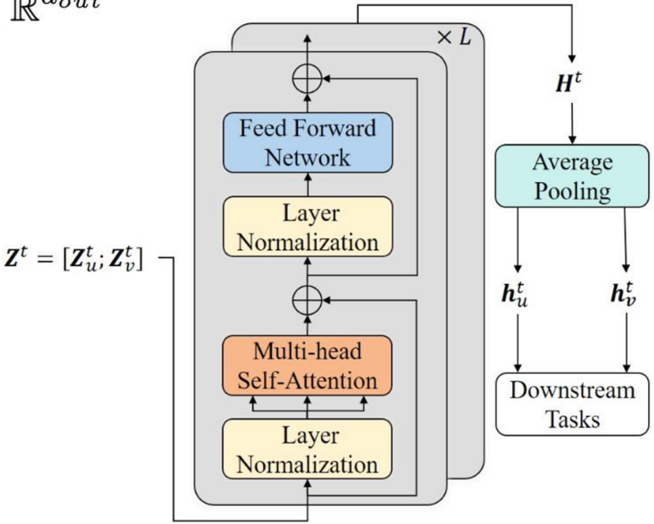
- Dividing the encoding sequence into multiple non-overlapping patches
- The patch size P, each patch is composed of P temporally adjacent interactions
  - Preservation of local temporal proximities via flattened patch encodings
  - Reduced computational cost by limiting the number of patches
- $X_{u,*}^t \in \mathbb{R}^{|S_u^t| \times d_*}$ will be divided into $l_u^t = \lceil \frac{|S_u^t|}{P} \rceil \rightarrow M_{u,*}^t \in \mathbb{R}^{l_u^t \times d_* \cdot P}$
- $\boldsymbol{Z}_{u,*}^t = \boldsymbol{M}_{u,*}^t \boldsymbol{W}_* + \boldsymbol{b}_* \in \mathbb{R}^{l_u^t \times d}$, $\boldsymbol{Z}_u^t = \boldsymbol{Z}_{u,N}^t \| \boldsymbol{Z}_{u,E}^t \| \boldsymbol{Z}_{u,T}^t \| \boldsymbol{Z}_{u,C}^t \in \mathbb{R}^{l_u^t \times 4d}$
- $\boldsymbol{Z}^t = [\boldsymbol{Z}_u^t; \boldsymbol{Z}_v^t] \in \mathbb{R}^{(l_u^t + l_v^t) \times 4d}$



Methodology

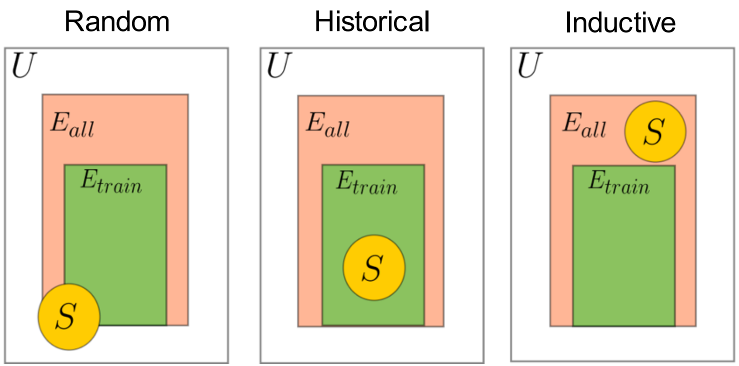
- Employing a Transformer Encoder
  - Input: $\boldsymbol{Z}^t = [\boldsymbol{Z}_u^t; \boldsymbol{Z}_v^t] \in \mathbb{R}^{(l_u^t + l_v^t) \times 4d}$
  - Aiming to learn the temporal dependencies within and across the sequences of u and v
  - Output: $\boldsymbol{h}_u^t = \mathrm{MEAN}\left(\boldsymbol{H}^t[: l_u^t, :]\right) \boldsymbol{W}_{out} + \boldsymbol{b}_{out} \in \mathbb{R}^{d_{out}},$

$$\boldsymbol{h}_v^t = \mathrm{MEAN}\left(\boldsymbol{H}^t[l_u^t : l_u^t + l_v^t, :]\right) \boldsymbol{W}_{out} + \boldsymbol{b}_{out} \in \mathbb{R}^{d_{out}}$$



Experiments

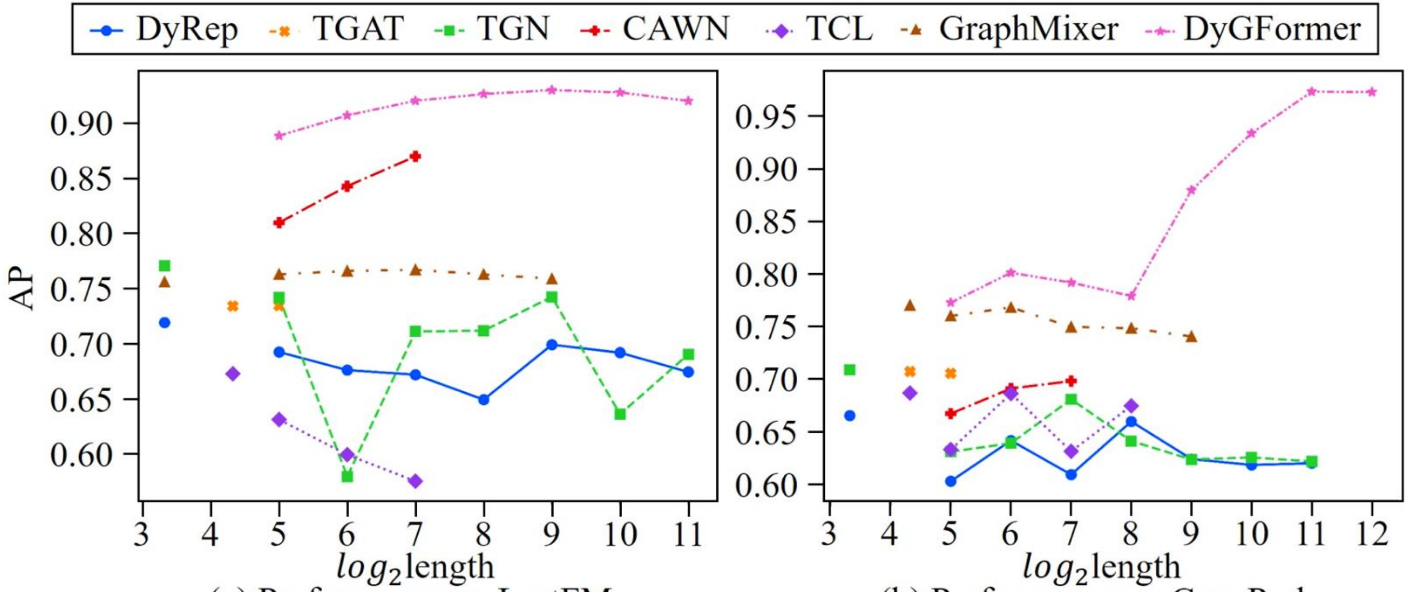
- Experimental Setup
  - Eight popular continuous-time dynamic graph learning baselines
    - Graph convolutions, memory networks, random walks and sequential models
  - Thirteen real-world dataset
    - Four bipartite graphs
    - Eight unipartite graphs
  - Link prediction
    - Transductive
    - Inductive
  - Various negative sampling strategies
    - Random
    - Historical
    - Inductive



Farimah Poursafaei, Andy Huang, Kellin Pelrine, and Reihaneh Rabbany. Towards better evaluation for dynamic link prediction
In Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2022.

Experiments

- Performance with Varying Input Lengths
  - Robust performance gains from leveraging longer histories via the patching technique



(a) Performance on LastFM. (b) Performance on Can. Parl..

Experiments

- Performance in AP for Transductive Setting

NSS	Datasets	JODIE	DyRep	TGAT	TGN	CAWN	EdgeBank	TCL	GraphMixer	DyGFormer
rnd	Wikipedia	96.50 ± 0.14	94.86 ± 0.06	96.94 ± 0.06	98.45 ± 0.06	98.76 ± 0.03	90.37 ± 0.00	96.47 ± 0.16	97.25 ± 0.03	**99.03 ± 0.02**
	Reddit	98.31 ± 0.14	98.22 ± 0.04	98.52 ± 0.02	98.63 ± 0.06	99.11 ± 0.01	94.86 ± 0.00	97.53 ± 0.02	97.31 ± 0.01	**99.22 ± 0.01**
	MOOC	80.23 ± 2.44	81.97 ± 0.49	85.84 ± 0.15	**89.15 ± 1.60**	80.15 ± 0.25	57.97 ± 0.00	82.38 ± 0.24	82.78 ± 0.15	87.52 ± 0.49
	LastFM	70.85 ± 2.13	71.92 ± 2.21	73.42 ± 0.21	77.07 ± 3.97	86.99 ± 0.06	79.29 ± 0.00	67.27 ± 2.16	75.61 ± 0.24	**93.00 ± 0.12**
	Enron	84.77 ± 0.30	82.38 ± 3.36	71.12 ± 0.97	86.53 ± 1.11	89.56 ± 0.09	83.53 ± 0.00	79.70 ± 0.71	82.25 ± 0.16	**92.47 ± 0.12**
	Social Evo.	89.89 ± 0.55	88.87 ± 0.30	93.16 ± 0.17	93.57 ± 0.17	84.96 ± 0.09	74.95 ± 0.00	93.13 ± 0.16	93.37 ± 0.07	**94.73 ± 0.01**
	UCI	89.43 ± 1.09	65.14 ± 2.30	79.63 ± 0.70	92.34 ± 1.04	95.18 ± 0.06	76.20 ± 0.00	89.57 ± 1.63	93.25 ± 0.57	**95.79 ± 0.17**
	Flights	95.60 ± 1.73	95.29 ± 0.72	94.03 ± 0.18	97.95 ± 0.14	98.51 ± 0.01	89.35 ± 0.00	91.23 ± 0.02	90.99 ± 0.05	**98.91 ± 0.01**
	Can. Parl.	69.26 ± 0.31	66.54 ± 2.76	70.73 ± 0.72	70.88 ± 2.34	69.82 ± 2.34	64.55 ± 0.00	68.67 ± 2.67	77.04 ± 0.46	**97.36 ± 0.45**
	US Legis.	75.05 ± 1.52	75.34 ± 0.39	68.52 ± 3.16	**75.99 ± 0.58**	70.58 ± 0.48	58.39 ± 0.00	69.59 ± 0.48	70.74 ± 1.02	71.11 ± 0.59
	UN Trade	64.94 ± 0.31	63.21 ± 0.93	61.47 ± 0.18	65.03 ± 1.37	65.39 ± 0.12	60.41 ± 0.00	62.21 ± 0.03	62.61 ± 0.27	**66.46 ± 1.29**
	UN Vote	63.91 ± 0.81	62.81 ± 0.80	52.21 ± 0.98	**65.72 ± 2.17**	52.84 ± 0.10	58.49 ± 0.00	51.90 ± 0.30	52.11 ± 0.16	55.55 ± 0.42
	Contact	95.31 ± 1.33	95.98 ± 0.15	96.28 ± 0.09	96.89 ± 0.56	90.26 ± 0.28	92.58 ± 0.00	92.44 ± 0.12	91.92 ± 0.03	**98.29 ± 0.01**
	Avg. Rank	5.08	5.85	5.69	2.54	4.31	7.54	6.92	5.46	**1.62**
hist	Wikipedia	83.01 ± 0.66	79.93 ± 0.56	87.38 ± 0.22	86.86 ± 0.33	71.21 ± 1.67	73.35 ± 0.00	89.05 ± 0.39	**90.90 ± 0.10**	82.23 ± 2.54
	Reddit	80.03 ± 0.36	79.83 ± 0.31	79.55 ± 0.20	81.22 ± 0.61	80.82 ± 0.45	73.59 ± 0.00	77.14 ± 0.16	78.44 ± 0.18	**81.57 ± 0.67**
	MOOC	78.94 ± 1.25	75.60 ± 1.12	82.19 ± 0.62	**87.06 ± 1.93**	74.05 ± 0.95	60.71 ± 0.00	77.06 ± 0.41	77.77 ± 0.92	85.85 ± 0.66
	LastFM	74.35 ± 3.81	74.92 ± 2.46	71.59 ± 0.24	76.87 ± 4.64	69.86 ± 0.43	73.03 ± 0.00	59.30 ± 2.31	72.47 ± 0.49	**81.57 ± 0.48**
	Enron	69.85 ± 2.70	71.19 ± 2.76	64.07 ± 1.05	73.91 ± 1.76	64.73 ± 0.36	76.53 ± 0.00	70.66 ± 0.39	**77.98 ± 0.92**	75.63 ± 0.73
	Social Evo.	87.44 ± 6.78	93.29 ± 0.43	95.01 ± 0.44	94.45 ± 0.56	85.53 ± 0.38	80.57 ± 0.00	94.74 ± 0.31	94.93 ± 0.31	**97.38 ± 0.14**
	UCI	75.24 ± 5.80	55.10 ± 3.14	68.27 ± 1.37	80.43 ± 2.12	65.30 ± 0.43	65.50 ± 0.00	80.25 ± 2.74	**84.11 ± 1.35**	82.17 ± 0.82
	Flights	66.48 ± 2.59	67.61 ± 0.99	**72.38 ± 0.18**	66.70 ± 1.64	64.72 ± 0.97	70.53 ± 0.00	70.68 ± 0.24	71.47 ± 0.26	66.59 ± 0.49
	Can. Parl.	51.79 ± 0.63	63.31 ± 1.23	67.13 ± 0.84	68.42 ± 3.07	66.53 ± 2.77	63.84 ± 0.00	65.93 ± 3.00	74.34 ± 0.87	**97.00 ± 0.31**
	US Legis.	51.71 ± 5.76	**86.88 ± 2.25**	62.14 ± 6.60	74.00 ± 7.57	68.82 ± 8.23	63.22 ± 0.00	80.53 ± 3.95	81.65 ± 1.02	85.30 ± 3.88
	UN Trade	61.39 ± 1.83	59.19 ± 1.07	55.74 ± 0.91	58.44 ± 5.51	55.71 ± 0.38	**81.32 ± 0.00**	55.90 ± 1.17	57.05 ± 1.22	64.41 ± 1.40
	UN Vote	70.02 ± 0.81	69.30 ± 1.12	52.96 ± 2.14	69.37 ± 3.93	51.26 ± 0.04	**84.89 ± 0.00**	52.30 ± 2.35	51.20 ± 1.60	60.84 ± 1.58
	Contact	95.31 ± 2.13	96.39 ± 0.20	96.05 ± 0.52	93.05 ± 2.35	84.16 ± 0.49	88.81 ± 0.00	93.86 ± 0.21	93.36 ± 0.41	**97.57 ± 0.06**
	Avg. Rank	5.46	5.08	5.08	3.85	7.54	5.92	5.46	4.00	**2.62**
ind	Wikipedia	75.65 ± 0.79	70.21 ± 1.58	87.00 ± 0.16	85.62 ± 0.44	74.06 ± 2.62	80.63 ± 0.00	86.76 ± 0.72	**88.59 ± 0.17**	78.29 ± 5.38
	Reddit	86.98 ± 0.16	86.30 ± 0.26	89.59 ± 0.24	88.10 ± 0.24	**91.67 ± 0.24**	85.48 ± 0.00	87.45 ± 0.29	85.26 ± 0.11	91.11 ± 0.40
	MOOC	65.23 ± 2.19	61.66 ± 0.95	75.95 ± 0.64	77.50 ± 2.91	73.51 ± 0.94	49.43 ± 0.00	74.65 ± 0.54	74.27 ± 0.92	**81.24 ± 0.69**
	LastFM	62.67 ± 4.49	64.41 ± 2.70	71.13 ± 0.17	65.95 ± 5.98	67.48 ± 0.77	**75.49 ± 0.00**	58.21 ± 0.89	68.12 ± 0.33	73.97 ± 0.50
	Enron	68.96 ± 0.98	67.79 ± 1.53	63.94 ± 1.36	70.89 ± 2.72	75.15 ± 0.58	73.89 ± 0.00	71.29 ± 0.32	75.01 ± 0.79	**77.41 ± 0.89**
	Social Evo.	89.82 ± 4.11	93.28 ± 0.48	94.84 ± 0.44	95.13 ± 0.56	88.32 ± 0.27	83.69 ± 0.00	94.90 ± 0.36	94.72 ± 0.33	**97.68 ± 0.10**
	UCI	65.99 ± 1.40	54.79 ± 1.76	68.67 ± 0.84	70.94 ± 0.71	64.61 ± 0.48	57.43 ± 0.00	76.01 ± 1.11	**80.10 ± 0.51**	72.25 ± 1.71
	Flights	69.07 ± 4.02	70.57 ± 1.82	75.48 ± 0.26	71.09 ± 2.72	69.18 ± 1.52	**81.08 ± 0.00**	74.62 ± 0.18	74.87 ± 0.21	70.92 ± 1.78
	Can. Parl.	48.42 ± 0.66	58.61 ± 0.86	68.82 ± 1.21	65.34 ± 2.87	67.75 ± 1.00	62.16 ± 0.00	65.85 ± 1.75	69.48 ± 0.63	**95.44 ± 0.57**
	US Legis.	50.27 ± 5.13	**83.44 ± 1.16**	61.91 ± 5.82	67.57 ± 6.47	65.81 ± 8.52	64.74 ± 0.00	78.15 ± 3.34	79.63 ± 0.84	81.25 ± 3.62
	UN Trade	60.42 ± 1.48	60.19 ± 1.24	60.61 ± 1.24	61.04 ± 6.01	62.54 ± 0.67	**72.97 ± 0.00**	61.06 ± 1.74	60.15 ± 1.29	55.79 ± 1.02
	UN Vote	**67.79 ± 1.46**	67.53 ± 1.98	52.89 ± 1.61	67.63 ± 2.67	52.19 ± 0.34	66.30 ± 0.00	50.62 ± 0.82	51.60 ± 0.73	51.91 ± 0.84
	Contact	93.43 ± 1.78	94.18 ± 0.10	94.35 ± 0.48	90.18 ± 3.28	89.31 ± 0.27	85.20 ± 0.00	91.35 ± 0.21	90.87 ± 0.35	**94.75 ± 0.28**
	Avg. Rank	6.62	6.38	4.15	4.38	5.46	5.62	4.69	4.46	**3.23**

Experiments

- Performance in AP for Inductive Setting

NSS	Datasets	JODIE	DyRep	TGAT	TGN	CAWN	TCL	GraphMixer	DyGFormer
rnd	Wikipedia	94.82 ± 0.20	92.43 ± 0.37	96.22 ± 0.07	97.83 ± 0.04	98.24 ± 0.03	96.22 ± 0.17	96.65 ± 0.02	**98.59 ± 0.03**
	Reddit	96.50 ± 0.13	96.09 ± 0.11	97.09 ± 0.04	97.50 ± 0.07	98.62 ± 0.01	94.09 ± 0.07	95.26 ± 0.02	**98.84 ± 0.02**
	MOOC	79.63 ± 1.92	81.07 ± 0.44	85.50 ± 0.19	**89.04 ± 1.17**	81.42 ± 0.24	80.60 ± 0.22	81.41 ± 0.21	86.96 ± 0.43
	LastFM	81.61 ± 3.82	83.02 ± 1.48	78.63 ± 0.31	81.45 ± 4.29	89.42 ± 0.07	73.53 ± 1.66	82.11 ± 0.42	**94.23 ± 0.09**
	Enron	80.72 ± 1.39	74.55 ± 3.95	67.05 ± 1.51	77.94 ± 1.02	86.35 ± 0.51	76.14 ± 0.79	75.88 ± 0.48	**89.76 ± 0.34**
	Social Evo.	91.96 ± 0.48	90.04 ± 0.47	91.41 ± 0.16	90.77 ± 0.86	79.94 ± 0.18	91.55 ± 0.09	91.86 ± 0.06	**93.14 ± 0.04**
	UCI	79.86 ± 1.48	57.48 ± 1.87	79.54 ± 0.48	88.12 ± 2.05	92.73 ± 0.06	87.36 ± 2.03	91.19 ± 0.42	**94.54 ± 0.12**
	Flights	94.74 ± 0.37	92.88 ± 0.73	88.73 ± 0.33	95.03 ± 0.60	97.06 ± 0.02	83.41 ± 0.07	83.03 ± 0.05	**97.79 ± 0.02**
	Can. Parl.	53.92 ± 0.94	54.02 ± 0.76	55.18 ± 0.79	54.10 ± 0.93	55.80 ± 0.69	54.30 ± 0.66	55.91 ± 0.82	**87.74 ± 0.71**
	US Legis.	54.93 ± 2.29	57.28 ± 0.71	51.00 ± 3.11	**58.63 ± 0.37**	53.17 ± 1.20	52.59 ± 0.97	50.71 ± 0.76	54.28 ± 2.87
	UN Trade	59.65 ± 0.77	57.02 ± 0.69	61.03 ± 0.18	58.31 ± 3.15	**65.24 ± 0.21**	62.21 ± 0.12	62.17 ± 0.31	64.55 ± 0.62
	UN Vote	56.64 ± 0.96	54.62 ± 2.22	52.24 ± 1.46	**58.85 ± 2.51**	49.94 ± 0.45	51.60 ± 0.97	50.68 ± 0.44	55.93 ± 0.39
	Contact	94.34 ± 1.45	92.18 ± 0.41	95.87 ± 0.11	93.82 ± 0.99	89.55 ± 0.30	91.11 ± 0.12	90.59 ± 0.05	**98.03 ± 0.02**
	Avg. Rank	4.69	5.77	5.23	3.77	3.77	5.77	5.23	**1.54**
hist	Wikipedia	68.69 ± 0.39	62.18 ± 1.27	84.17 ± 0.22	81.76 ± 0.32	67.27 ± 1.63	82.20 ± 2.18	**87.60 ± 0.30**	71.42 ± 4.43
	Reddit	62.34 ± 0.54	61.60 ± 0.72	63.47 ± 0.36	64.85 ± 0.85	63.67 ± 0.41	60.83 ± 0.25	64.50 ± 0.26	**65.37 ± 0.60**
	MOOC	63.22 ± 1.55	62.93 ± 1.24	76.73 ± 0.29	77.07 ± 3.41	74.68 ± 0.68	74.27 ± 0.53	74.00 ± 0.97	**80.82 ± 0.30**
	LastFM	70.39 ± 4.31	71.45 ± 1.76	76.27 ± 0.25	66.65 ± 6.11	71.33 ± 0.47	65.78 ± 0.65	**76.42 ± 0.22**	76.35 ± 0.52
	Enron	65.86 ± 3.71	62.08 ± 2.27	61.40 ± 1.31	62.91 ± 1.16	60.70 ± 0.36	67.11 ± 0.62	**72.37 ± 1.37**	67.07 ± 0.62
	Social Evo.	88.51 ± 0.87	88.72 ± 1.10	93.97 ± 0.54	90.66 ± 1.62	79.83 ± 0.38	94.10 ± 0.31	94.01 ± 0.47	**96.82 ± 0.16**
	UCI	63.11 ± 2.27	52.47 ± 2.06	70.52 ± 0.93	70.78 ± 0.78	64.54 ± 0.47	76.71 ± 1.00	**81.66 ± 0.49**	72.13 ± 1.87
	Flights	61.01 ± 1.65	62.83 ± 1.31	64.72 ± 0.36	59.31 ± 1.43	56.82 ± 0.57	64.50 ± 0.25	**65.28 ± 0.24**	57.11 ± 0.21
	Can. Parl.	52.60 ± 0.88	52.28 ± 0.31	56.72 ± 0.47	54.42 ± 0.77	57.14 ± 0.07	55.71 ± 0.74	55.84 ± 0.73	**87.40 ± 0.85**
	US Legis.	52.94 ± 2.11	**62.10 ± 1.41**	51.83 ± 3.95	61.18 ± 1.10	55.56 ± 1.71	53.87 ± 1.41	52.03 ± 1.02	56.31 ± 3.46
	UN Trade	55.46 ± 1.19	55.49 ± 0.84	55.28 ± 0.71	52.80 ± 3.19	55.00 ± 0.38	**55.76 ± 1.03**	54.94 ± 0.97	53.20 ± 1.07
	UN Vote	61.04 ± 1.30	60.22 ± 1.78	53.05 ± 3.10	**63.74 ± 3.00**	47.98 ± 0.84	54.19 ± 2.17	48.09 ± 0.43	52.63 ± 1.26
	Contact	90.42 ± 2.34	89.22 ± 0.66	**94.15 ± 0.45**	88.13 ± 1.50	74.20 ± 0.80	90.44 ± 0.17	89.91 ± 0.36	93.56 ± 0.52
	Avg. Rank	5.38	5.46	4.00	4.54	5.92	3.92	3.54	**3.23**
ind	Wikipedia	68.70 ± 0.39	62.19 ± 1.28	84.17 ± 0.22	81.77 ± 0.32	67.24 ± 1.63	82.20 ± 2.18	**87.60 ± 0.29**	71.42 ± 4.43
	Reddit	62.32 ± 0.54	61.58 ± 0.72	63.40 ± 0.36	64.84 ± 0.84	63.65 ± 0.41	60.81 ± 0.26	64.49 ± 0.25	**65.35 ± 0.60**
	MOOC	63.22 ± 1.55	62.92 ± 1.24	76.72 ± 0.30	77.07 ± 3.40	74.69 ± 0.68	74.28 ± 0.53	73.99 ± 0.97	**80.82 ± 0.30**
	LastFM	70.39 ± 4.31	71.45 ± 1.75	76.28 ± 0.25	69.46 ± 4.65	71.33 ± 0.47	65.78 ± 0.65	**76.42 ± 0.22**	76.35 ± 0.52
	Enron	65.86 ± 3.71	62.08 ± 2.27	61.40 ± 1.30	62.90 ± 1.16	60.72 ± 0.36	67.11 ± 0.62	**72.37 ± 1.38**	67.07 ± 0.62
	Social Evo.	88.51 ± 0.87	88.72 ± 1.10	93.97 ± 0.54	90.65 ± 1.62	79.83 ± 0.39	94.10 ± 0.32	94.01 ± 0.47	**96.82 ± 0.17**
	UCI	63.16 ± 2.27	52.47 ± 2.09	70.49 ± 0.93	70.73 ± 0.79	64.54 ± 0.47	76.65 ± 0.99	**81.64 ± 0.49**	72.13 ± 1.86
	Flights	61.01 ± 1.66	62.83 ± 1.31	64.72 ± 0.37	59.32 ± 1.45	56.82 ± 0.56	64.50 ± 0.25	**65.29 ± 0.24**	57.11 ± 0.20
	Can. Parl.	52.58 ± 0.86	52.24 ± 0.28	56.46 ± 0.50	54.18 ± 0.73	57.06 ± 0.08	55.46 ± 0.69	55.76 ± 0.65	**87.22 ± 0.82**
	US Legis.	52.94 ± 2.11	**62.10 ± 1.41**	51.83 ± 3.95	61.18 ± 1.10	55.56 ± 1.71	53.87 ± 1.41	52.03 ± 1.02	56.31 ± 3.46
	UN Trade	55.43 ± 1.20	55.42 ± 0.87	55.58 ± 0.68	52.80 ± 3.24	54.97 ± 0.38	**55.66 ± 0.98**	54.88 ± 1.01	52.56 ± 1.70
	UN Vote	61.17 ± 1.33	60.29 ± 1.79	53.08 ± 3.10	**63.71 ± 2.97**	48.01 ± 0.82	54.13 ± 2.16	48.10 ± 0.40	52.61 ± 1.25
	Contact	90.43 ± 2.33	89.22 ± 0.65	**94.14 ± 0.45**	88.12 ± 1.50	74.19 ± 0.81	90.43 ± 0.17	89.91 ± 0.36	93.55 ± 0.52
	Avg. Rank	5.31	5.54	3.85	4.38	5.85	3.92	3.46	**3.31**

Experiments

- DyGFormer vs DyGFormer w/o Neighbor Co-occurrence Encoding
  - Changed Link Ratio
    - The ratio of common neighbors in source node's sequence and destination node's sequence
    - $|S_u \cap S_v| / |S_u \cup S_v|$
  - Common Neighbor Ratio
    - The ratio of the changed links to their original set
    - $|\mathrm{FN}\rightarrow\mathrm{TP}|/|\mathrm{FN}|$, $|\mathrm{FP}\rightarrow\mathrm{TN}|/|\mathrm{FP}|$, $|\mathrm{TP}\rightarrow\mathrm{FN}|/|\mathrm{TP}|$, and $|\mathrm{TN}\rightarrow\mathrm{FP}|/|\mathrm{TN}|$

Datasets	CLR (%)				CNR (%)			
	FN→TP	FP→TN	TP→FN	TN→FP	FN→TP	FP→TN	TP→FN	TN→FP
Wikipedia	68.36	72.73	1.68	1.69	18.16	0.01	0.10	2.49
UCI	71.45	94.11	7.29	1.82	19.08	2.49	3.35	13.02
Flights	83.66	83.83	1.73	2.11	37.09	2.28	7.06	20.28
US Legis.	31.63	23.67	6.63	1.59	69.92	62.13	61.14	63.80
UN Vote	44.02	36.46	28.95	30.53	78.57	81.39	80.86	77.02

Experiments

- When Will DyGFormer Be a Good Choice?
  - Link Ratio
    - The ratio of links in their corresponding positive or negative set
  - Superiority on datasets with high CNR for positive links and low CNR for negative links
  - Performance variation caused by increased CNR of negative samples in hist/ind settings

TP/(TP+FN) TN/(TN+FP)

Datasets	LR (%)		CNR (%)	
	TP	TN	TP	TN
Wikipedia	92.74	97.19	59.09	0.01
UCI	82.70	96.77	28.03	1.45
Flights	96.13	95.33	47.58	1.40
US Legis.	78.95	56.83	75.18	53.98
UN Vote	65.18	45.43	56.24	76.02

FP/(TN+FP)

Datasets	LR(%)			CNR(%)		
	rnd	hist	ind	rnd	hist	ind
Wikipedia	2.81	89.28	94.53	0.02	14.00	11.66
UCI	3.23	64.93	76.42	9.98	12.22	13.81
Flights	4.67	94.52	92.94	0.01	35.62	30.29
US Legis.	43.17	17.31	21.21	79.40	87.51	75.51
UN Vote	54.57	39.90	52.92	79.60	75.53	79.15

Conclusion

- Existing Models
  - Node embedding computation without modeling node correlations
  - Interaction truncation for computational feasibility
- A neighbor co-occurrence encoding
  - Exploiting the correlations of nodes
- A patching technique with Transformer
  - Improved capture of long-term temporal dependencies
- Performance Variability Across Datasets
  - Suitability for datasets with high CNR of true positives

Thank you